

NANYANG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS
Higher 1

CANDIDATE
NAME

CLASS

BIOLOGY

Paper 1 Multiple Choice

8875/01

September 2014

1 hour

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name and CT on the Answer Sheet in the spaces provided unless this has been done for you.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

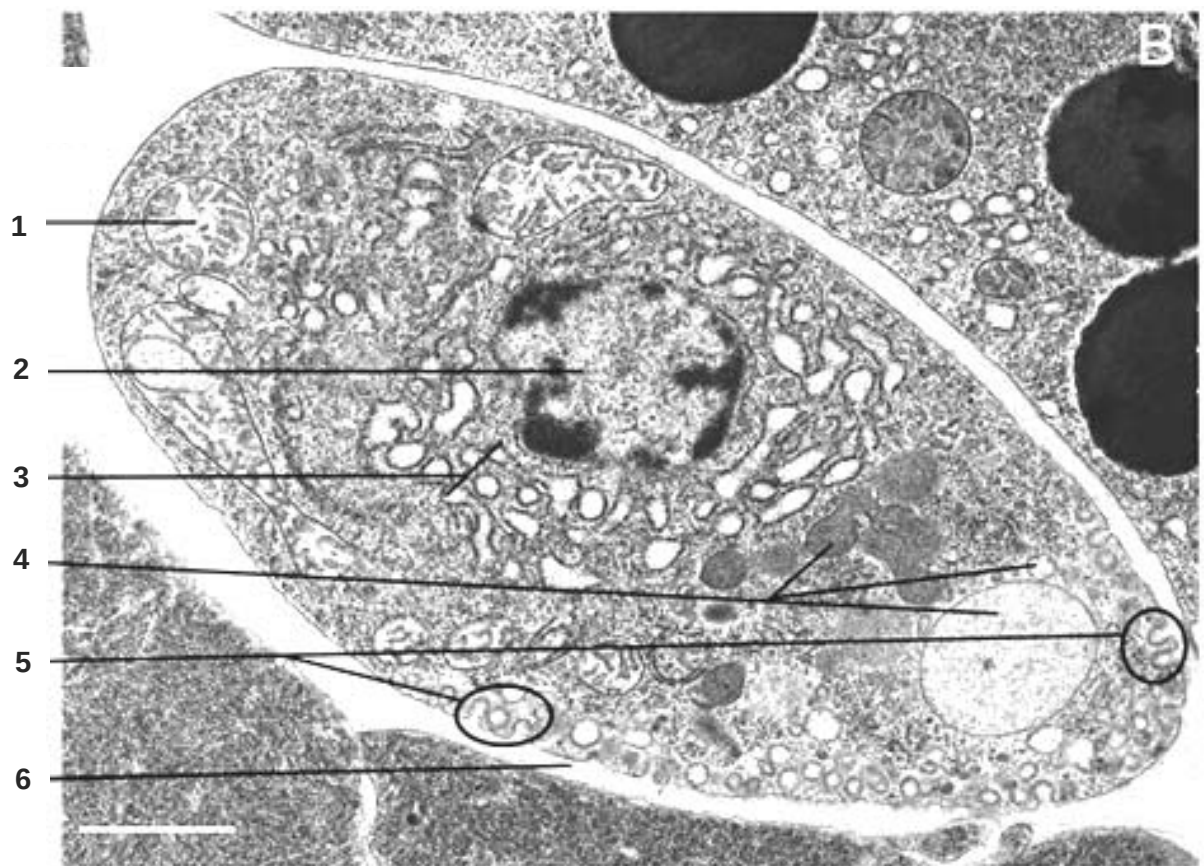
Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

Calculators may be used.

This document consists of **13** printed pages and **1** blank page.

[Turn over



- 1 Which of the structures in the above electron micrograph are found abundantly in actively respiring tissues?
 - A 4 only
 - B 1 only**
 - C 2 and 5 only
 - D 3 and 6 only

- 2 Which of the following statements about the cell is true?
 - A Endocytosis is an active process because energy is required to move materials into the cell against their concentration gradients.
 - B In simple diffusion, materials enter the cell down their concentration gradients whereas facilitated diffusion can move materials against their concentration gradients.
 - C In osmosis, water, ions and small polar molecules enter the cell by diffusing through a selectively permeable membrane down their concentration gradients.
 - D Materials such as ions and small polar molecules can enter the cell against their concentration gradient by active transport.**

- 3 Which of the following best accounts for the difference in structure between amylose which is helical and cellulose which is fibrous?
- A Presence or absence of 180° rotation of alternate subunits.
 - B Presence or absence of branching in the macromolecule
 - C Subunits are either monosaccharides or amino acids.
 - D Different tendency of macromolecule to form hydrogen bonds with water.
- 4 The flowchart below shows part of an enzyme-catalysed sequence which involves 4 compounds (**W** – **Z**) and 3 enzymes (**E1** – **E3**).

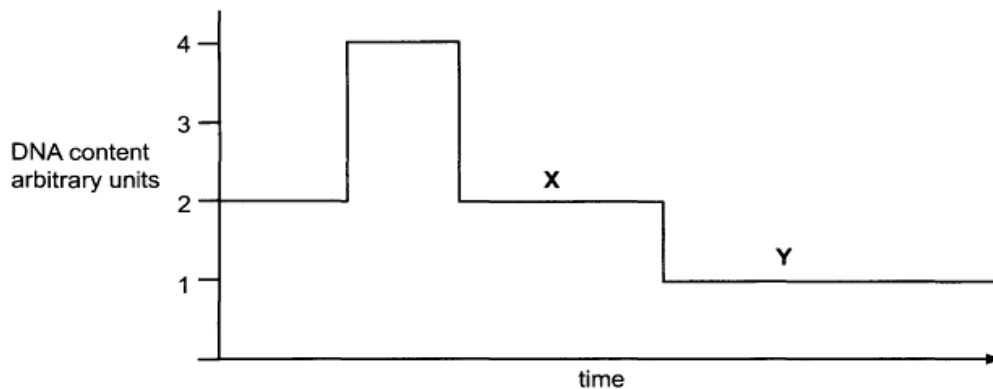


The addition of another compound **M** results in no change in the concentration of compound **W**, an accumulation of compound **X**, and a near absence of both compound **Y** and compound **Z**.

What does this information indicate about the mode of action of compound **M**, and which is a method by which the production of compound **Z** can be increased in the presence of compound **M**?

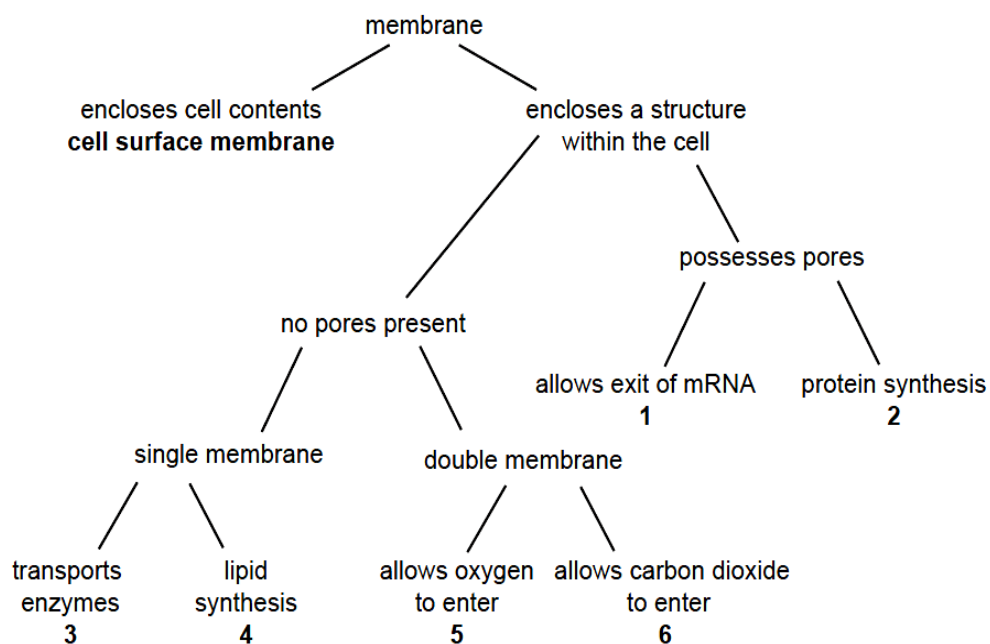
	Mode of action of compound M	Method to increase production of compound Z
A	It inhibits E1 .	Increase concentration of compound X in the reaction mixture.
B	It activates E1 .	Increase concentration of E2 in the reaction mixture.
C	It inhibits E2 .	Increase concentration of compound Y in the reaction mixture.
D	It inhibits E3 .	Increase concentration of compound Y in the reaction mixture.

- 5 The graph shows the amount of DNA present in the nuclei of cells undergoing division in a multicellular organism.



	Ploidy of cell at end of X	Ploidy of cell at end of Y
A	diploid	diploid
B	diploid	haploid
C	haploid	haploid
D	haploid	diploid

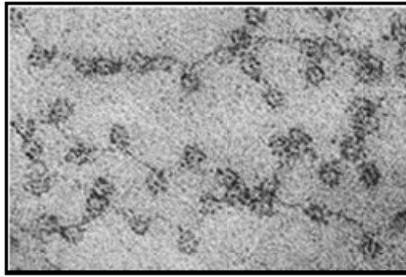
- 6 Membranes within and at the surface of cells have different roles. The flowchart shows a scheme to identify various organelles within the cell, by describing the structure and function of their membranes.



Which of the following outcomes correctly identifies the various cellular organelles?

	I	II	III	IV	V	VI
A	chloroplast	vesicle	smooth ER	rough ER	nucleolus	mitochondrion
B	nucleolus	rough ER	vesicle	smooth ER	nucleus	mitochondrion
C	nucleus	smooth ER	mitochondrion	rough ER	vesicle	chloroplast
D	nucleus	rough ER	vesicle	smooth ER	mitochondrion	chloroplast

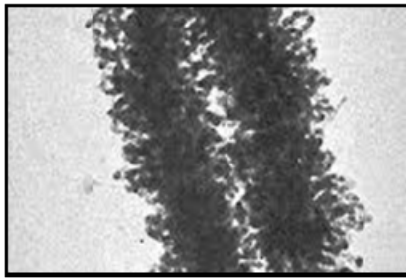
- 7 If there were 30 amino acids and DNA contained two types of nitrogenous bases, what would be the minimum number of bases per codon that could code for proteins?
- A 3
B 5
 C 6
 D 8
- 8 The electron micrographs I to IV show various levels of structural organisation of DNA



I



II



III



IV

Which of the following best describes the significance of these structures at each level of packing?

	I	II	III	IV
A	To allow further compaction of DNA	To allow DNA to be replicated with high fidelity	To allow equal distribution of genetic material	To allow some DNA to be transcribed
B	To allow some DNA to be transcribed	To allow DNA to be replicated with high fidelity	To allow equal distribution of genetic material	To allow further compaction of DNA
C	To allow further compaction of DNA	To allow equal distribution of genetic material	To allow DNA to be replicated with high fidelity	To allow some DNA to be transcribed
D	To allow some DNA to be transcribed	To allow equal distribution of genetic material	To allow further compaction of DNA	To allow DNA to be replicated with high fidelity

- 9 A student obtained a sample of DNA. mRNA was transcribed from this DNA and the two samples were subsequently purified. He then separated the two strands of the DNA sample.

The base compositions of each strand and that of the mRNA were analysed. The results of the analysis are shown in the table below.

	A	G	C	T	U
DNA strand 1	19.1	26.0	31.0	23.9	0.0
DNA strand 2	24.2	30.8	25.7	19.3	0.0
DNA strand 3	20.5	25.2	29.8	24.5	0.0
mRNA	19.0	25.9	30.8	0.0	24.3

Which strand of DNA is the coding strand, serving as a template for mRNA synthesis?

- A Strand 1
B Strand 2
 C Strand 3
 D Strand 2 and 3
- 10 One individual, organism **P**, in a sexually-reproducing population of mammals has two forms of a protein in its heart muscle. The normal form is identical to that in most other members of the population, while the variant form differs by one amino acid. The diagram shows the amino acid sequences in part of the normal protein and in the same part of the variant protein.

Normal protein	Variant protein
-gly-lys-lys-gly-glu-	-glu-lys-gly-gly-glu-

What could account for the presence of the normal protein and the variant protein in P?

- The sequence of amino acids was altered in a cell in the heart muscle of P's mother.
 - The sequence of amino acids was altered in a cell in the ovaries of P's mother.
 - The sequence of bases was altered in the DNA in a cell in the heart muscle of P.
 - The sequence of bases was altered in the DNA in a cell in an ovary of P.
 - The sequence of bases was altered in the DNA in a cell in a testis of P's grandfather.
- A 1, 2 and 4 only
 B 1 and 3 only
 C 2, 4 and 5 only
D 3 and 5 only

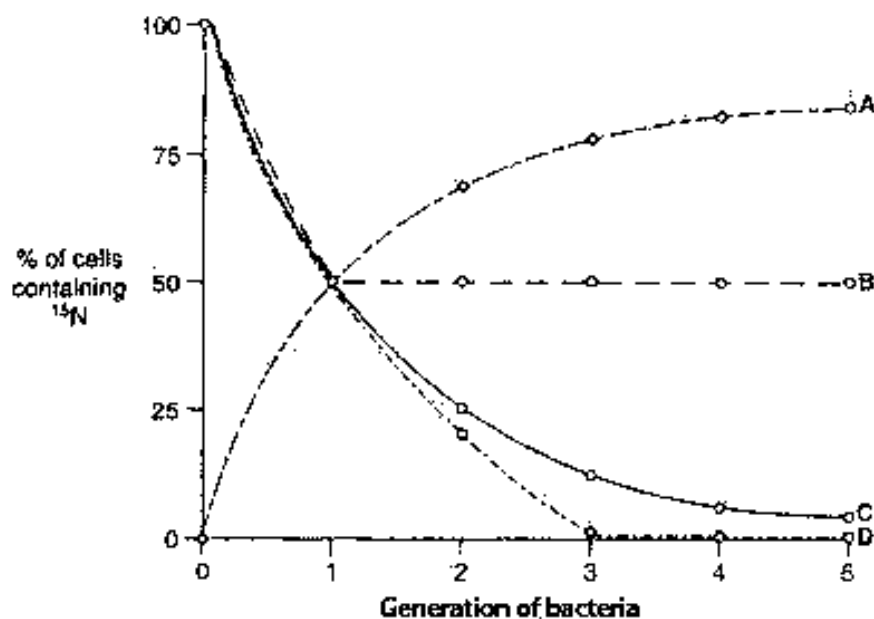
- 11 Genetic drugs are short sequences of nucleotides. One example is an anti-sense drug made from RNA nucleotides that are complementary to the protein-coding mRNA.

Which is the process that is affected by the anti-sense drug and what is its most probable mode of action?

- A It prevents DNA replication by binding to the anti-sense strand of DNA that codes for the disease-causing protein.
- B It prevents transcription by binding to the sense strand of DNA that codes for the disease-causing protein.
- C It prevents translation by binding to the mRNA that contains information for the disease-causing protein.
- D It prevents the normal functioning of disease-related protein when it is translated into a protein that inhibits the disease-causing protein.

- 12 Bacteria were cultured in a medium containing heavy nitrogen (^{15}N) until all the DNA was labelled. These bacteria were then grown in a medium containing only normal nitrogen (^{14}N) for five generations. The percentage of cells containing ^{15}N in each generation was estimated.

Which curve provides evidence that DNA replication is semi conservative? CCCCCCCC

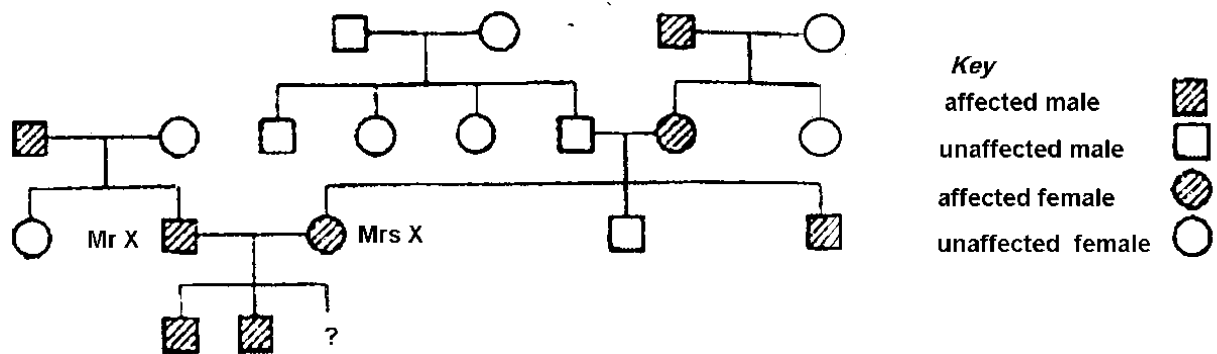


- 13 A polypeptide has the amino acid sequence glycine – arginine – lysine – serine. The table below gives possible tRNA anticodons for each amino acid.

amino acid	tRNA anticodons
arginine	UCC GCG
glycine	CCA CCU
lysine	UUC UUU
serine	AGG UCG

- A CCACGCAAGAGC
 B CCTTCCTTCTCG
 C GGAAGGAAAAGC
 D GGTTGGTTGTGC

- 14 The family tree below was assembled by a genetic counsellor for Mr and Mrs X who suffer from heart disease due to hypercholesterolaemia. Children who inherit the dominant mutant allele from both parents rarely survive beyond puberty.



What is the probability that Mr and Mrs X's third child will be unaffected?

- A 0.25
 B 0.50
 C 0.75
 D 1.00
- 15 In insects, sex is determined by a ZW chromosome system. Males are ZZ and females are ZW. A lethal recessive allele is sometimes present on the Z chromosome in moths.

What would be the sex ratio in the offspring of a cross between a male heterozygote and a normal female?

- A 1 male : 1 female
 B 2 male : 1 female
 C 1 male : 2 female
 D 3 male : 1 female

- 16 Fruit flies (*Drosophila*), homozygous for long wings, were crossed with flies homozygous for vestigial wings. The F₁ and F₂ generations were raised at three different temperatures. At each temperature, the F₁ generation all had long wings.

The table shows the results in the F₂ generation.

Temperature	Result
21°C	¾ long wings, ¼ vestigial wings
26 °C	¾ long wings, ¼ intermediate wing length
31 °C	All long wings

Which statement explains these results?

- A Wing length is under polygenic control.
 - B Long wing and vestigial wing illustrate codominance at 26 °C.
 - C Long wing is dominant at higher temperatures but vestigial wing is dominant at lower temperatures.
 - D Vestigial wing is recessive but causes a vestigial wing phenotype only at lower temperatures.
- 17 The table below shows the number of chromosomes in the parent species of the cabbage *Brassica oleracea*, the carrot *Raphanus sativus* as well as the various intermediate hybrids.

type of cell	number of chromosomes per cell
Parent cabbage	18
Parent carrot	18
Parent's gamete	9
F ₁ hybrid	18
F ₁ gamete	18
F ₂ hybrid	36
F ₂ gamete	18
F ₃ hybrid	36

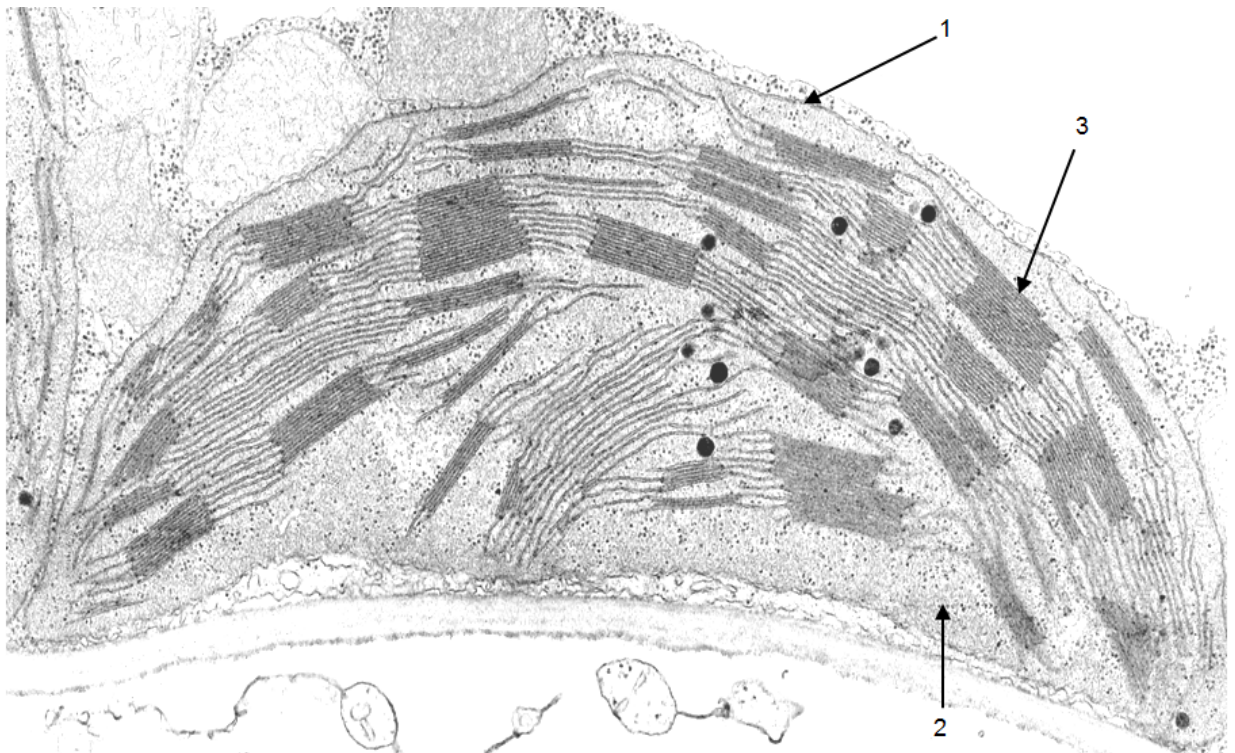
Chromosomal mutation takes place during the

- A Fusion of parents' gametes
- B Formation of F₁ gametes
- C Fusion of F₁ gametes
- D Formation of F₂ gametes

- 18** A metabolic poison rotenone was added to a mixture of homogenized cells with intact chloroplasts. It is discovered that rotenone can bind irreversibly to the electrons carriers on the thylakoid membrane in chloroplasts and alter their conformational shape.

In what way would the Calvin cycle be affected by rotenone poisoning?

- A** Calvin cycle will stop, as the accumulation of triose phosphate will create feedback inhibition on glucose formation.
 - B** The rate of Calvin cycle will decrease due to the accumulation of glycerate-3-phosphate.
 - C** The rate of Calvin cycle will increase since electrons can flow directly to final electron acceptor to form reduced NADPH.
 - D** There will be no effect since Calvin cycle is an enzyme-controlled process.
- 19** Which of the following is correctly associated with the function or characteristic of the labelled structure?



	Structure containing electron carrier	Structure where sugar is made	Structure with a double membrane
A	3	1	2
B	1	3	2
C	3	2	1
D	2	1	3

- 20 The table shows events that may occur in the Calvin cycle, the Krebs cycle or in glycolysis. In which of these processes do the events occur?

	ATP used	dehydrogenation	reduced NADP used
A	Calvin cycle	glycolysis	Krebs cycle
B	glycolysis	Calvin cycle	Krebs cycle
C	glycolysis	Krebs cycle	Calvin cycle
D	Krebs cycle	glycolysis	Calvin cycle

- 21 Lactic acid is produced whenever muscle tissue contracts. Muscle tissue can contract in the absence of oxygen and lactic acid accumulates until fatigue sets in. On return to aerobic conditions, the muscle tissue can contract again and the lactic acid is metabolised.

Which of the following is the best interpretation of these observations?

- A Formation of lactic acid is oxygen-dependent.
 - B In anaerobic conditions, lactic acid is produced faster than it can be removed.
 - C Lactic acid is produced only in anaerobic conditions.
 - D Muscle contraction is independent of oxygen availability.
- 22 Potassium cyanide is known to interfere with the formation and use of ATP in cell metabolism.

If the use of potassium cyanide resulted in an accumulation of a solute in a cell, it may be deduced that the solute normally enters by

- A active transport.
 - B diffusion.
 - C pinocytosis.
 - D reverse osmosis.
- 23 An interbreeding population of finches became separated geographically, forming two isolated groups. Each group then became subject to different selective pressures. One group was then introduced into the habitat of the other.

Which one of the following would determine whether they now formed two distinct species?

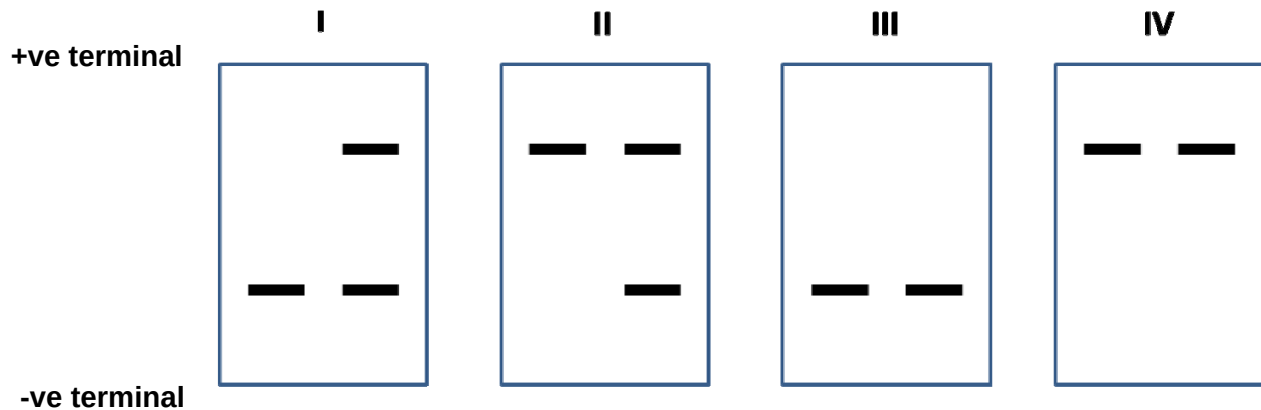
- A They had been separated for more than three million years.
- B They failed to produce fertile F_1 hybrids.
- C They showed marked differences in the shape of their beaks.
- D Several genes now possessed different base sequences.

- 24 Which of the following ideas was **not** part of Charles Darwin's theory of evolution by natural selection?
- A Organisms produced more offspring than the environment can support.
 - B Variation between individuals arises by gene mutation.
 - C Only those individuals best adapted to the environment survive and reproduce.
 - D Adaptive variation is inherited.
- 25 The binomial nomenclature system devised by Carolus Linnaeus specified that each animal and plant should be named according to their
- A class and order
 - B family and genus
 - C family and order
 - D genus and species
- 26 The restriction enzyme *DdeI* recognizes a sequence on DNA, CTNAG (where N is any base). A portion of the β -hemoglobin sequence encoding normal hemoglobin is CCTGAGGAG.
- If the mutation leading to sickle cell anemia had been CCTCAGGAG, would you be able to distinguish the two alleles with a restriction digest?
- A Yes, the normal allele will be digested, the sickle allele will not.
 - B Yes, the sickle allele will be digested, the normal allele will not.
 - C No, both will be digested by *DdeI*.
 - D No, neither will be digested by *DdeI*.
- 27 A plasmid has two antibiotic resistance genes, one for ampicillin and one for tetracycline. It is treated with a restriction enzyme that cuts in the middle of the ampicillin gene. DNA fragments containing a human globin gene were cut with the same enzyme. The plasmids and fragments are mixed, treated with ligase, and used to transform bacterial cells. Clones that have taken up the recombinant DNA are the ones that
- A are blue and can grow on plates with both antibiotics.
 - B can grow on plates with ampicillin but not with tetracycline.
 - C can grow on plates with tetracycline but not with ampicillin.
 - D cannot grow with any antibiotics.

- 28** Cystic fibrosis (CF) is an autosomal recessive genetic disorder. An individual must have two copies of the mutated CFTR gene to express the disease phenotype. One of the most common CF-causing mutation resulted in a loss of phenylalanine located at position 508 of the protein.

The DNA sequence of the CF locus from the offspring of 2 carriers are removed and separated by gel electrophoresis.

Which pattern of bands corresponds to two of the offspring that are phenotypically normal?



- A I only
 B II only
 C I and III
 D II and IV
- 29** What are some similarities between all cancer cells and all stem cells?
- A They replicate indefinitely, are non-differentiated and can be found in various parts of the body.
 B They replicate indefinitely, lack cell-cell adhesion and are able to move from one location in the body to another.
 C They lack contact inhibition, are non-differentiated and are regulated by molecular signals.
 D They are able to move from one location in the body to another, are regulated by molecular signals and can be found in various parts of the body.
- 30** Which of the following is **not** a concern with growing genetically engineered crops *Bt* corn?
- A Flow of the Bt gene from cultivated corn to wild relative thereby enhancing the weediness of certain corn relatives, which could then lead to loss of biodiversity.
 B Transfer of marker genes that convey antibiotics resistance, used in the identification of successfully transformed cells, to bacteria that resides in the gut of humans thereby rendering the bacteria antibiotic resistant.
 C Toxic effects of Bt on non-target lepidopteran insects (e.g. monarch butterfly larvae of the insect order *Lepidoptera*) could upset ecological balance.
 D Transfer of Bt genes to the pests thereby increasing their resistance to the Bt toxin.